

Northwind Industrial

HX-400 quench-water duty - design note

Revision C, 20 January 2026. Prepared for the reliability team.

Why the unit is smaller than the datasheet asked for

The tendered fouling resistance was copied from a table and applied to a fluid described in two words. Sizing to it buys area, lowers the clean velocity, and makes the unit foul faster than the model assumed.

Sizing instead to the cleaning interval the site actually holds gives a shell two sizes smaller at 412 square metres, and a longer run.

Basis

Design pressure 42 barg. Design temperature 285 C.

Tube side duplex 2205. Shell side carbon steel, 3 mm allowance.

TEMA AES, one shell pass, two tube passes.

Delivery

Twenty-four weeks from drawing approval, on the reserved mill slot.

